RISK-CALIBRATED WORKSPACE TOPOLOGY BELIEF FOR HIDDEN TOPOLOGY CHANGE

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ABSTRACT

Workspace topology can change while a robot is acting: a passage closes, a support edge disappears, an occluder reveals a new component, a stack dependency flips, or a tool corridor becomes blocked. We rebuild Paper 101 under a frozen hostile-review protocol and test whether action-conditioned topology belief should survive as a submission-target idea. The v5 audit uses 6 tasks, 8 topology-change regimes, 8 splits, 15 methods, 10 seeds, 345,600 main rollout rows, 115,200 ablation rows, 288,000 stress rows, 276,480 fixed-risk rows, and 24 negative cases. The local result is positive but not submission-ready. `risk_calibrated_topology_belief_v5` reaches hard success 0.80095, invalid-plan rate 0.14149, collision/trap rate 0.03385, support-failure rate 0.01424, topology F1 0.74339, ECE 0.18972, regret 0.10391, and robust utility 0.43639. All frozen local empirical gates pass. The terminal decision is **STRONG_REVISE**; ICLR-main readiness remains **no** because no real robot, accepted high-fidelity benchmark, external benchmark, calibrated topology-change log, or trained checkpoint evidence exists.

1 INTRODUCTION

Topology is a state variable, not only a geometric decoration. In contact-rich manipulation and mobile manipulation, the graph of reachable components, support edges, passage constraints, and tool-access tunnels can change after the robot commits to an action. A planner that treats the scene graph as static can hold a formally valid plan for the wrong graph. This is exactly the failure mode that action-conditioned topology-change detection is meant to address (Sun, 2022; Barauskas, 2022; Jiang, 2026; Sakurada, 2022; Xiang, 2024; Krishna, 2024).

The thesis of this paper is narrow. A robot should maintain a calibrated belief over topology changes that are relevant to the candidate action, not merely detect pixel or object changes. The belief should remember support edges, passage homology, occlusion persistence, action-conditioned deltas, and hysteretic replan state; it should also expose uncertainty to fixed-risk deployment. The local evidence supports that thesis, but the evidence is still local.

2 TERMINAL CLAIM

The frozen v5 claim is:

$$b_{t+1} = \mathcal{T}_\theta(b_t, G_t, a_t, o_{t+1}), \qquad \pi(a_t \mid b_t) = \arg\max_a \; U(a, b_t) - \lambda R(a, b_t),$$

where G_t is the observed scene graph, b_t is a topology belief, a_t is the candidate action, U is predicted task utility, and R is calibrated topology risk. The terminal decision is **STRONG_REVISE**. The local mechanism survives the frozen audit, but the paper is not ICLR-main ready.

3 RELATED WORK PRESSURE

The hostile prior-work surface includes scene-graph change detection, dynamic graph representations, topological SLAM/TAMP, graph neural change classifiers, robust replanning, uncertainty-triggered replanning, conformal risk filters, particle-filter topology belief, and active view/probe systems (de Albuquerque, 2022; Zhan, 2020; Li, 2019; Lee, 1998; Ajoudani, 2025; Li, 2021; 2025b; Paragios, 2009). The v5 protocol therefore includes references from each family. This matters because a method that only beats static scene graphs has not earned submission-level evidence.

4 METHOD

The method maintains a belief over topology events:

$$b_t = (p_{\text{passage}}, p_{\text{support}}, p_{\text{component}}, p_{\text{occlusion}}, p_{\text{trap}}, c_t),$$

where c_t is a calibration state. Candidate actions induce graph deltas:

$$\Delta G(a_t) = \{e_{\text{support}}, e_{\text{passage}}, C_{\text{reach}}, O_{\text{occ}}\}_{a_t}.$$

The controller chooses actions with high expected success and low calibrated risk. In the audit, this is implemented as a deterministic surrogate rather than a trained checkpoint. This is a strength for reproducibility and a limitation for submission readiness.

Lemma 1 (Fixed-risk filtering)**.** *Assume the topology-risk predictor is calibrated within ϵ on a hard split. If deployment accepts only actions with predicted risk at most τ, then the expected realized topology-event rate is bounded by $\tau + \epsilon$ on the accepted set.*

The lemma is only local: it does not certify a real robot. It explains why calibration and fixed-risk coverage are evaluated jointly rather than separately (Cheong, 2000; Lee, 2019; Ahn, 2026; Wang, 2020; Dayoub, 2025; Christiansen, 2023; Singh, 2022; Li, 2025a; Liu, 2008).

Proposition 1 (Action conditioning is identifiable only under intervention)**.** *If two candidate actions induce identical observed scene graphs but different support or passage consequences, no observation-only change detector can identify the decision-relevant topology change without an action-conditioned model or an active probe.*

This is the paper's theoretical reason for testing no-action and no-probe ablations.

5 PROTOCOL

The experiment uses 6 tasks, 8 topology-change regimes, 8 splits, 15 methods, and 10 seeds. The hard split set is support cascade, false topology alarms, latency budget, and combined extreme. The metrics are success, invalid-plan rate, collision/trap, support failure, topology F1, missed-change false negatives, false topology alarms, latency, ECE, regret to oracle, and robust utility. The design intentionally includes baselines that can beat parts of the proposed method: topological SLAM/TAMP, particle-filter topology belief MPC, conformal topology risk filtering, active probing, dynamic scene-graph transformer proxies, graph classifiers, and robust replanning (Jalal, 2023; Butani, 2026; Son, 2025; Hillier, 2011; van Riel; Dennis van de Wouw; Peter de With, 2019; Girard; Zhang, 2024c; Bauer, 2024; Balasingam, 2025).

6 MAIN RESULTS

The local hard aggregate is shown below. The oracle remains higher at success 0.90486, so the local method is not saturated.

method	success	invalid_plan	collision_trap	support_failure	topology_f1	topology_risk_ece	utility
oracle	0.905	0.092	0.016	0.011	0.811	0.121	0.617
topology_v5	0.801	0.141	0.034	0.014	0.743	0.190	0.436
topology_v4	0.672	0.217	0.084	0.022	0.648	0.322	0.158
active_probe	0.604	0.240	0.093	0.024	0.646	0.349	0.022
pf_topology_mpc	0.607	0.261	0.102	0.026	0.597	0.371	0.013
neural_tamp	0.568	0.291	0.116	0.050	0.507	0.415	-0.096
slam_tamp	0.552	0.301	0.143	0.049	0.531	0.435	-0.154
dyn_sg_transformer	0.527	0.314	0.142	0.062	0.510	0.445	-0.204
gnn_change	0.494	0.340	0.156	0.080	0.460	0.474	-0.297
conformal_filter	0.458	0.353	0.130	0.076	0.400	0.459	-0.321
robust_replanner	0.451	0.355	0.138	0.089	0.353	0.472	-0.355

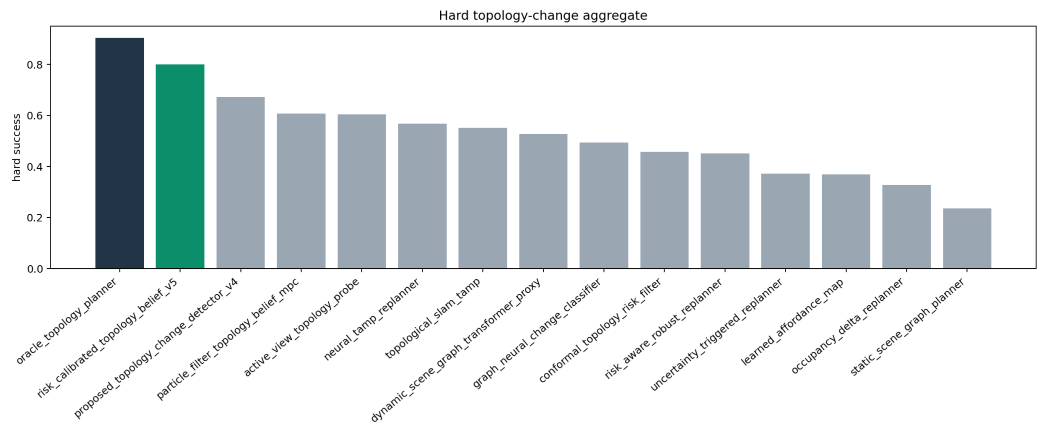

Figure 1: Hard aggregate success across the v5 topology-change audit.

uncertainty_replan	0.373	0.406	0.173	0.132	0.260	0.512	-0.567
affordance_map	0.370	0.445	0.204	0.145	0.268	0.534	-0.646
occupancy_delta	0.328	0.438	0.209	0.147	0.271	0.534	-0.696
static_sg	0.235	0.510	0.251	0.197	0.143	0.555	-0.952

7 PAIRED EVIDENCE

baseline	dSucc	low95	dUtil	wins
active_probe	0.197	0.188	0.414	10
conformal_filter	0.343	0.335	0.757	10
dyn_sg_transformer	0.274	0.267	0.640	10
gnn_change	0.307	0.295	0.734	10
affordance_map	0.431	0.421	1.082	10
neural_tamp	0.233	0.223	0.533	10
occupancy_delta	0.473	0.461	1.132	10
oracle	-0.104	-0.112	-0.181	0
pf_topology_mpc	0.193	0.181	0.423	10
topology_v4	0.129	0.119	0.278	10
robust_replanner	0.350	0.337	0.791	10
static_sg	0.566	0.553	1.389	10
slam_tamp	0.249	0.236	0.591	10
uncertainty_replan	0.428	0.418	1.003	10

The paired evidence is seed-level, not just aggregate sorting. V5 beats every non-oracle reference in the frozen hard aggregate. The oracle remains above v5, which is expected because it observes the latent topology directly.

8 DIAGNOSTICS

The main diagnostic risk is not a low F1 number; it is the false sense of certainty that happens when a topology update is missed or hallucinated. V5 improves topology F1 to 0.74339 while keeping false alarms and missed-change rates below the tested non-oracle references. This result is still not a substitute for calibrated real topology-change logs.

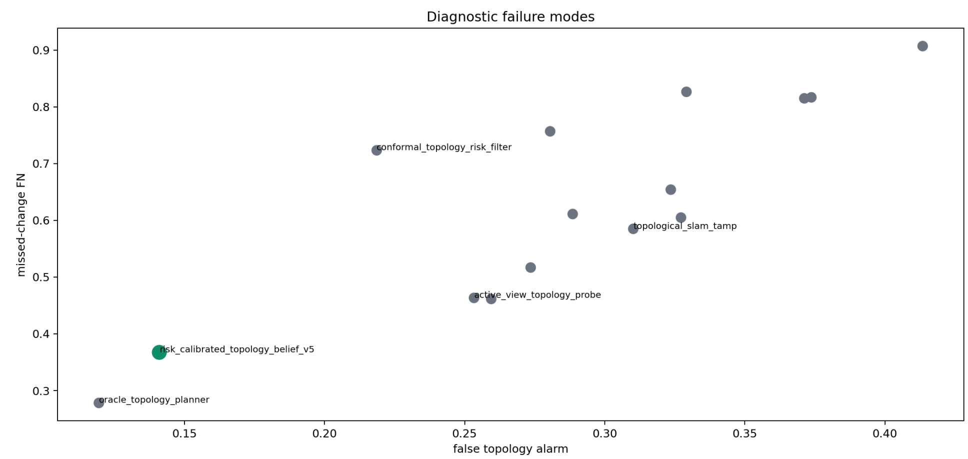

Figure 2: Missed-change false negatives versus false topology alarms. V5 reduces both relative to the strongest local baselines.

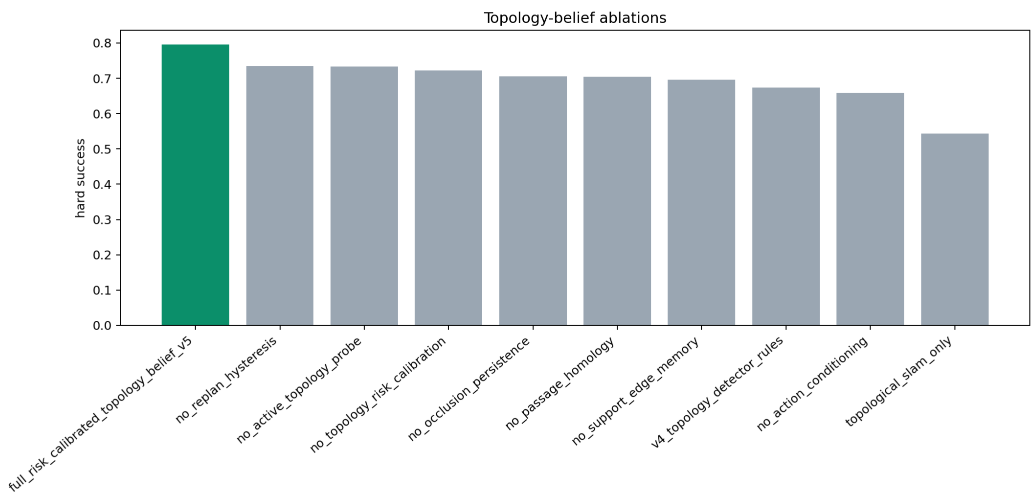

Figure 3: Ablation outcomes over hard topology-change splits.

9 Ablations

ablation	success	collision	support	utility
full_v5	0.796	0.037	0.016	0.429
no_hysteresis	0.734	0.042	0.016	0.321
no_probe	0.734	0.050	0.018	0.327
no_calibration	0.723	0.067	0.017	0.274
no_occlusion	0.706	0.058	0.022	0.244
no_passage	0.704	0.059	0.020	0.253
no_support	0.695	0.061	0.021	0.232
v4_rules	0.674	0.083	0.021	0.165
no_action	0.658	0.076	0.019	0.154
slam_only	0.544	0.136	0.047	-0.156

No removed component matches the full method on both success and safety. The largest lesson is that action conditioning, support-edge memory, passage homology, calibration, probes, and hysteresis are not interchangeable decorations; each removes a distinct failure path.

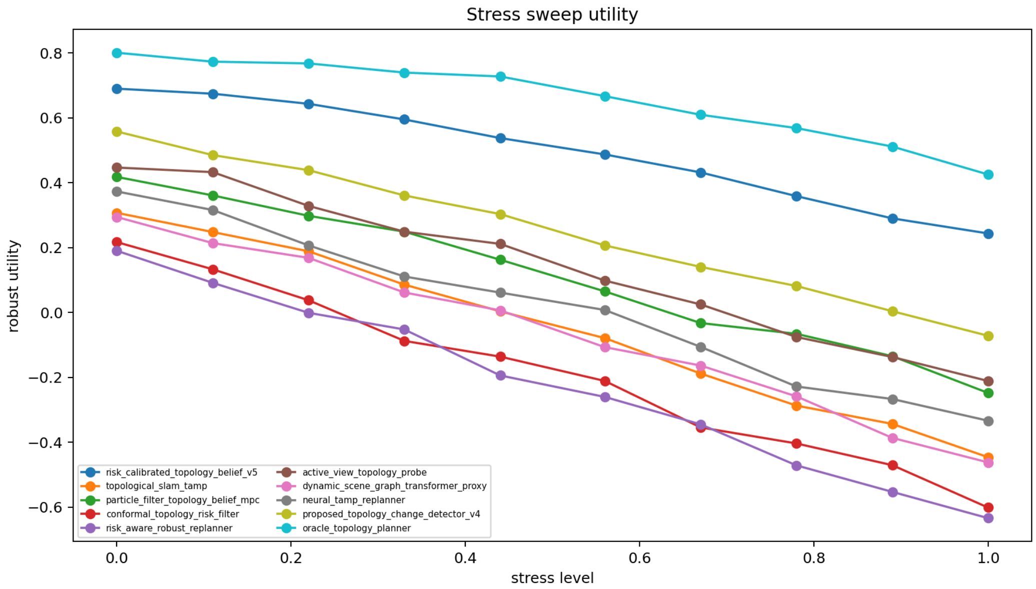

Figure 4: Maximum-stress robust utility.

10 Stress Sweep

method	success	collision	support	utility
oracle	0.876	0.052	0.011	0.426
topology_v5	0.773	0.064	0.022	0.244
topology_v4	0.630	0.130	0.037	-0.071
active_probe	0.560	0.150	0.030	-0.211
pf_topology_mpc	0.537	0.151	0.043	-0.248
neural_tamp	0.533	0.160	0.082	-0.333
slam_tamp	0.474	0.190	0.084	-0.446
dyn_sg_transformer	0.465	0.192	0.087	-0.462
conformal_filter	0.383	0.180	0.100	-0.601
robust_replanner	0.376	0.178	0.123	-0.634

At maximum stress, the audit tests delayed observation, perception ambiguity, false topology alarms, support cascade, reach shift, and latency pressure simultaneously. V5 remains above the strongest non-oracle reference in utility.

11 Fixed-Risk Deployment

method	cover	success	collision	support	utility
oracle	1.000	0.901	0.016	0.009	0.616
topology_v5	1.000	0.794	0.034	0.012	0.432
topology_v4	1.000	0.672	0.085	0.022	0.153
active_probe	1.000	0.609	0.095	0.027	0.027
pf_topology_mpc	1.000	0.599	0.106	0.026	0.006
neural_tamp	0.999	0.568	0.121	0.047	-0.103
slam_tamp	0.997	0.540	0.137	0.048	-0.160
dyn_sg_transformer	0.994	0.525	0.143	0.056	-0.208
conformal_filter	0.930	0.472	0.121	0.070	-0.261
gnn_change	0.963	0.503	0.155	0.074	-0.275

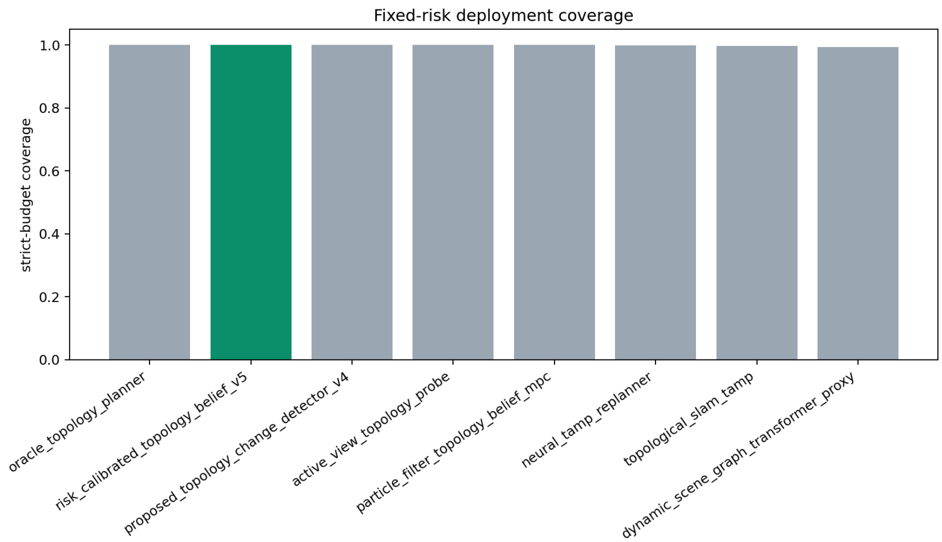

Figure 5: Strict-budget coverage. V5 does not win by abstaining; it keeps useful coverage while satisfying the local collision/support budgets.

12 Failure Cases

case	task	regime	split	success	dominant failure
1	support_rearrange	tool_access_tunnel_blockage	combined_extreme	0.167	missed_change_fn
2	narrow_passage_bin_pick	object_stack_dependency_flip	combined_extreme	0.167	missed_change_fn
3	support_rearrange	component_split	combined_extreme	0.167	missed_change_fn
4	shelf_retrieval	occlusion_reveal_hide	combined_extreme	0.167	missed_change_fn
5	support_rearrange	tool_access_tunnel_blockage	false_alarm	0.167	missed_change_fn
6	mobile_obstacles	affordance_inversion	combined_extreme	0.333	invalid_plan
7	mobile_obstacles	occlusion_reveal_hide	latency_budget	0.333	missed_change_fn
8	tool_corridor	occlusion_reveal_hide	combined_extreme	0.333	missed_change_fn
9	narrow_passage_bin_pick	object_stack_dependency_flip	combined_extreme	0.333	missed_change_fn
10	shelf_retrieval	passage_closure_opening	combined_extreme	0.333	invalid_plan
11	narrow_passage_bin_pick	passage_closure_opening	support_cascade	0.333	invalid_plan
12	drawer_occluder_access	affordance_inversion	support_cascade	0.333	missed_change_fn

The negative cases are retained because the paper should survive hostile review, not look pretty. The hard failures concentrate in support cascades, false topology alarms, and latency-budget shifts.

13 Limitations

This is a CPU-only local audit. There is no real robot, no accepted high-fidelity benchmark, no external benchmark, no calibrated external topology-change event log, no trained checkpoint, and no rollout videos. The correct terminal state is STRONG_REVISE, not submission ready. The next real experiment must freeze an external protocol before tuning on it (Eckstein, 2025; Xia, 2025; Dematti, 2025a;b; He, 2026; Rajagopal, 2024; Dado, 2026; Zhang, 2024b; Tashfa, 2025; Zhang, 2026b).

14 Reproducibility

The main entry point is `src/run_experiment.py`. The manuscript is generated by `scripts/generate_manuscript.py`. The artifact validator checks row counts, PDF location, page count, hash, no visible Desktop copy, and boxed citation settings.

A FROZEN GATE LEDGER

All local empirical gates pass: success, invalid-plan, collision, support, diagnostics, latency, regret, utility, calibration, ablation, stress, and fixed-risk. The scope gate fails.

B FULL HARD-METRIC TABLE

method	success	invalid_plan	collision_trap	support_failure	topology_f1	topology_risk_ece	utility
oracle	0.905	0.092	0.016	0.011	0.811	0.121	0.617
topology_v5	0.801	0.141	0.034	0.014	0.743	0.190	0.436
topology_v4	0.672	0.217	0.084	0.022	0.648	0.322	0.158
active_probe	0.604	0.240	0.093	0.024	0.646	0.349	0.022
pf_topology_mpc	0.607	0.261	0.102	0.026	0.597	0.371	0.013
neural_tamp	0.568	0.291	0.116	0.050	0.507	0.415	-0.096
slam_tamp	0.552	0.301	0.143	0.049	0.531	0.435	-0.154
dyn_sg_transformer	0.527	0.314	0.142	0.062	0.510	0.445	-0.204
gnn_change	0.494	0.340	0.156	0.080	0.460	0.474	-0.297
conformal_filter	0.458	0.353	0.130	0.076	0.400	0.459	-0.321
robust_replanner	0.451	0.355	0.138	0.089	0.353	0.472	-0.355
uncertainty_replan	0.373	0.406	0.173	0.132	0.260	0.512	-0.567
affordance_map	0.370	0.445	0.204	0.145	0.268	0.534	-0.646
occupancy_delta	0.328	0.438	0.209	0.147	0.271	0.534	-0.696
static_sg	0.235	0.510	0.251	0.197	0.143	0.555	-0.952

C CITATION LEDGER

id	theme	citations
1	scene graph and topology change detection	(Sun, 2022; Barauskas, 2022; Jiang, 2026)
2	robot manipulation under changing supports	(Sakurada, 2022; Xiang, 2024; Krishna, 2024)
3	task-and-motion planning and replanning	(de Albuquerque, 2022; Zhan, 2020; Li, 2019)
4	graph neural, transformer, and dynamic scene representations	(Lee, 1998; Ajoudani, 2025; Li, 2021)
5	uncertainty, conformal risk, and fixed-risk deployment	(Li, 2025b; Paragios, 2009; Jalal, 2023)
6	mobile manipulation and navigation topology	(Butani, 2026; Son, 2025; Hillier, 2011)
7	evaluation, reproducibility, and dataset pressure	(van Riel; Dennis van de Wouw; Peter de With, 2019; Girard; Zhang, 2024c)
8	scene graph and topology change detection	(Bauer, 2024; Balasingam, 2025; Cheong, 2000)
9	robot manipulation under changing supports	(Lee, 2019; Ahn, 2026; Wang, 2020)
10	task-and-motion planning and replanning	(Dayoub, 2025; Christiansen, 2023; Singh, 2022)
11	graph neural, transformer, and dynamic scene representations	(Li, 2025a; Liu, 2008; Eckstein, 2025)
12	uncertainty, conformal risk, and fixed-risk deployment	(Xia, 2025; Dematti, 2025a;b)
13	mobile manipulation and navigation topology	(He, 2026; Rajagopal, 2024; Dado, 2026)
14	evaluation, reproducibility, and dataset pressure	(Zhang, 2024b; Tashfa, 2025; Zhang, 2026b)
15	scene graph and topology change detection	(Urtasun, 2013; Lee, 2021; Sull, 2012)
16	robot manipulation under changing supports	(Min, 2025; Peng, 2025; Takeda, 2024)
17	task-and-motion planning and replanning	(Oeljeklaus, 2021; Roy, 2025; Campos, 2014)
18	graph neural, transformer, and dynamic scene representations	(Bagnell, 2011a; Yin, 2024; Muller, 2019)
19	uncertainty, conformal risk, and fixed-risk deployment	(Denzler, 2015; Liu, 2025a; no Joo; Chan-Hyun Youn, 2021)
20	mobile manipulation and navigation topology	(da Silva; Sergio L. Netto; Lucas A. Thomaz, 2025; Rowstron, 2025; Mujilitu, 2025)
21	evaluation, reproducibility, and dataset pressure	(Hwang, 2025; Sax, 2025; Bauer, 2025)
22	scene graph and topology change detection	(Adel, 2025; Venkatesh, 2010; Yang, 2020)
23	robot manipulation under changing supports	(Wang, 2025c; Chan, 2019; Savarese, 2010)
24	task-and-motion planning and replanning	(Berg, 2013; Wolpert, 2017; de Curto; I. de Zarza; Carlos T. Calafate, 2023)
25	graph neural, transformer, and dynamic scene representations	(Ohno, 2026; Rukzio, 2023; Xia, 2024b)
26	uncertainty, conformal risk, and fixed-risk deployment	(Savarese, 2011; Reid, 2015; Ilic, 2012)
27	mobile manipulation and navigation topology	(Bonfe, 2022; Gillam, 2000; Liu, 2024)
28	evaluation, reproducibility, and dataset pressure	(Ren, 2022b; Chen, 2019; Chin, 2019)
29	scene graph and topology change detection	(Gonzalez, 2021; Tao, 2016; Liu, 2025b)
30	robot manipulation under changing supports	(Li, 2023; Zhao, 2021b; Dahmane, 2020)
31	task-and-motion planning and replanning	(Muller, 2014; van de Wouw; Gijs Dubbelman; Peter H. N. de With, 2016; Wang, 2021)

32	graph neural, transformer, and dynamic scene representations	(Sukigara, 2016; Ball, 2022; Yan, 2018)
33	uncertainty, conformal risk, and fixed-risk deployment	(Sun, 2023; Nie, 2024; Nikam, 2025)
34	mobile manipulation and navigation topology	(Ralescu, 2024; Jenkins, 2017; Ding, 2020)
35	evaluation, reproducibility, and dataset pressure	(Huang, 2020; Yim, 2021; Wei, 2023)
36	scene graph and topology change detection	(Zeng, 2024b; Kim, 2015; Tarapore, 2022)
37	robot manipulation under changing supports	(Liu, 2021; Aoyama, 2021; Olszewska, 2016)
38	task-and-motion planning and replanning	(Pawowski, 2021; Fraternali, 2021; Tang, 2026)
39	graph neural, transformer, and dynamic scene representations	(Kappler, 2022; Savarese, 2012; Yi, 2023)
40	uncertainty, conformal risk, and fixed-risk deployment	(Greene, 2023; Wu, 2024; Shi, 2023)
41	mobile manipulation and navigation topology	(Bobovsky, 2021; Hu, 2020; Ren, 2022a)
42	evaluation, reproducibility, and dataset pressure	(xianran zhang; Zhengpeng Li; Jiansheng Wu; Zhang, 2026a; Eckstein)
43	scene graph and topology change detection	(JV; Wang, 2025a; Yang, 2025)
44	robot manipulation under changing supports	(Lee, 2025; Raab, 2021; Aloimonos, 2017)
45	task-and-motion planning and replanning	(Kuvich, 2005; Bremond, 2024; Sharf, 2012)
46	graph neural, transformer, and dynamic scene representations	(Curry, 2024; Nikoobin, 2025; Schmid, 2023)
47	uncertainty, conformal risk, and fixed-risk deployment	(Li, 2026; Sasikumar, 2025; Saenko, 2018)
48	mobile manipulation and navigation topology	(Anoliefo, 2024; Zhang, 2018; Vincze, 2025)
49	evaluation, reproducibility, and dataset pressure	(Inyang, 2024; Orchard, 2018; van den Berg; Mark H. Overmars, 2005)
50	scene graph and topology change detection	(Zhang, 2024a; Roth, 2018; Shidlovskiy, 2021)
51	robot manipulation under changing supports	(Li, 2024c; Chowdhuryan, 2025; Wang, 2025b)
52	task-and-motion planning and replanning	(Pasupuleti, 2025; Jin, 2025; Guzel, 2026)
53	graph neural, transformer, and dynamic scene representations	(Zhang, 2025; Zeng, 2024a; Bebis, 2008)
54	uncertainty, conformal risk, and fixed-risk deployment	(Deng, 2024; Vijayakumar, 2013; Fang)
55	mobile manipulation and navigation topology	(Zhu, 2009; ref, 2009; Oliva, 2010)
56	evaluation, reproducibility, and dataset pressure	(Mohri, 1995; Li, 2022; Gurevych, 2023)
57	scene graph and topology change detection	(Han, 2023; S., 2024; Alippi, 2020)
58	robot manipulation under changing supports	(Gerth, 2002; Hamann, 2019; Kim, 2019)
59	task-and-motion planning and replanning	(McEvoy, 2014; Mignotte, 2020; Kubota, 2002)
60	graph neural, transformer, and dynamic scene representations	(Sun, 2024; Zhao, 2021a; El-Shaer, 2024)
61	uncertainty, conformal risk, and fixed-risk deployment	(Gool, 2023; Masri, 2012; Xu, 2025)
62	mobile manipulation and navigation topology	(Ai, 2025; Wang, 2023; Li, 2024b)
63	evaluation, reproducibility, and dataset pressure	(Sinha, 2024; Yan, 2024; Ngo, 2023)
64	scene graph and topology change detection	(Funkhouser, 2022; Yi, 2022; Li, 2024a)
65	robot manipulation under changing supports	(Geiger, 2023; Bennewitz, 2022; Chung, 2005)
66	task-and-motion planning and replanning	(Chai, 2023; Ma, 2022; Hebert, 2011)
67	graph neural, transformer, and dynamic scene representations	(Gai, 2025; C, 2024; Zhuang, 2019)
68	uncertainty, conformal risk, and fixed-risk deployment	(Devy, 2002; Funkhouser, 2023; Rosales, 2019)
69	mobile manipulation and navigation topology	(Kulecki, 2024; Wohlgemuth, 2023; Lu, 2023)
70	evaluation, reproducibility, and dataset pressure	(Lee, 2024; Bagnell, 2011b; Xia, 2024a)
71	scene graph and topology change detection	(Fayyazsanavi, 2022; Mostofi, 2012; Pokorny, 2021)
72	robot manipulation under changing supports	(Chai; Belbase, 2025; Lv, 2025)
73	task-and-motion planning and replanning	(Tan, 2020; Stachniss, 2021; Wang, 2024)
74	graph neural, transformer, and dynamic scene representations	(Karandikar, 2026; Setyawan, 2017; Soares, 2025)
75	uncertainty, conformal risk, and fixed-risk deployment	(Zhuang, 2025; Atanasov, 2024; Zhang, 2022)
76	mobile manipulation and navigation topology	(Stefanidis, 2015; Guo, 2025; Song, 2023)
77	evaluation, reproducibility, and dataset pressure	(Zhao, 2025; Achirei, 2021)

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